Mohamed Badr

Software Engineer | Alexandria, Egypt
(+20) 1210082277 | mohamedbadrsaad23@gmail.com | GitHub | LinkedIn | Portfolio

EDUCATION

Graduated 7/2023 **B.Sc in Electronics and Communication Engineering**,
Faculty of Engineering - Alexandria University.

- **GPA: 3.55/4 - Department Rank: 6th**
- **Graduation project:** self-driving car system using deep learning and V2V communication, mastering data collection, deep learning model training, hardware integration, and performance tuning
- **Courses:** OS, Data structures and algorithms, Programming, Probability and statistics, Network
- **Languages:** Arabic-Native • English-Advanced • German-Basic

WORK EXPERIENCE

Freelance Fullstack Developer | Remote 6/2024 - Present

- Built and deployed full-stack web apps for using Next.js, React.js, Node.js, MongoDB, Tailwind CSS.
- Delivered responsive, modern interfaces and robust RESTful APIs.

Instrumentation and Control Engineer | Rashpetco (Shell JV) 2/2025 - Present

- Part of Rashpetco's digitalization team, contributing to internal applications that improve engineering workflows and reduce manual effort.
- Contributed to digitalization projects including the Deferral Management System, Instruments Management System, and Metering Best Match Finder.
- Troubleshot and analyzed control logic in distributed, safety-critical systems (DCS/ESD)
- Worked with industrial communication protocols and data-driven diagnostics

Software Developer | SEITech Solutions (OJT) 2/2024 – 5/2024

- Gained experience with version control systems, SDLC processes, and software design patterns
- Utilized DOORS for efficient requirements management in software engineering projects

COURSES

Internship at National Technology Institute (NTI) Full Stack Development track

- Technical training in full-stack development and in soft skills and project management
- Developed web applications using HTML, CSS/SASS, JavaScript, React.js, Node.js, and RESTful API

Professional Web Development by Udacity

- HTML - CSS – SASS – JavaScript - RESTful API - NodeJS

PROJECTS

- **Shell-JV Deferral Management System** *(Next.js, PostgreSQL, Better Auth)***:** web app for managing company deferrals through multi-level approval workflows, centralized tracking, and faster follow-up.
- **Talié E-Commerce Platform** *(Next.js 16, Tailwind CSS, Prisma ORM, PostgreSQL/Supabase, Paymob):* Built a modern fashion e-commerce app with product catalog, collections, cart, wishlist, customer accounts, order tracking, admin dashboard, inventory handling, and Paymob payment integration.
- **RASHPETCO Instruments Management System** *(React.Js, Node.js, MongoDB, Shadcn)*: Full-stack app for managing calibration workflows, instruments, and engineers with role-based auth, PDF reports
- **Metering Best Match Finder** *(JavaScript, Dynamic Programming)**:*** Tool to compute optimal input combinations to reach target values for client use cases
- **BadrPicture** *(React.js, TypeScript, Shadcn, Appwrite, Vite)***:** Modern social app with native mobile feel, infinite scroll, and optimized performance
- **Responsive Recipe Finder** *(HTML, CSS, JS)**:*** Meal information app with recipes, preparation videos

Technical Skills

- **Web Development:** Node.js, Next.js, Express.js, MySQL, MongoDB, Appwrite, HTML, CSS/SASS, React.js, Bootstrap, Material-UI, Shadcn, Three.js
- **Programming Languages:** TypeScript, JavaScript, C, C++, Python
- **Software Development:** OOP, Data structures, Algorithms, Design Patterns
- **DevOps & Tools:** Git, Visual Studio, Jupyter Notebook, Linux, Vite, CI/CD (GitHub Actions)
- **SDLC:** Waterfall, V-Model, Agile methodologies

Interpersonal Skills

- Problem-solving - Quick learner - Autonomous - Teamwork – Adaptability - Proactive

Activities

- **Team Coordinator – InnovEgypt:** worked on multiple projects emphasizing design thinking
- **Organizer – Student Union:** Organizer at Newcomers' submission and medical examination